Probabilistic Interpolation

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Interpolation

> Spatial interpolation is the process of predicting values of a target variable or attribute at an unobserved location, from a set of values at known (observed) locations

Deterministic vs. Probabalistic Interpolation

- **Deterministic-** interpolated values calculated as a function of distance
- **Probabilistic-** interpolated values are *predicted* based on statistical models

Kriging

- Previous methods predict z as a (weighted) function of distance
- Treat the observations as perfect (no error)
- If we imagine that z is the outcome of some spatial process such that:

$$z(\mathbf{x}) = \mu(\mathbf{x}) + \epsilon(\mathbf{x})$$

then any observed value of z is some function of the process ($\mu(\mathbf{x})$) and some error ($\epsilon(\mathbf{x})$)

- Kriging exploits autocorrelation in $\epsilon(\mathbf{x})$ to identify the trend and interpolate accordingly

Autocorrelation

- **Correlation** the tendency for two variables to be related
- **Autocorrelation** the tendency for observations that are closer (in space or time) to be correlated
- **Positive autocorrelation** neighboring observations have ϵ with the same sign
- **Negative autocorrelation** neighboring observations have ϵ with a different sign (rare in geography)

Ordinary Kriging

- Assumes that the deterministic part of the process ($\mu(\mathbf{x})$) is an unknown constant (μ)

$$z(\mathbf{x}) = \mu + \epsilon(\mathbf{x})$$

Steps for Ordinary Kriging

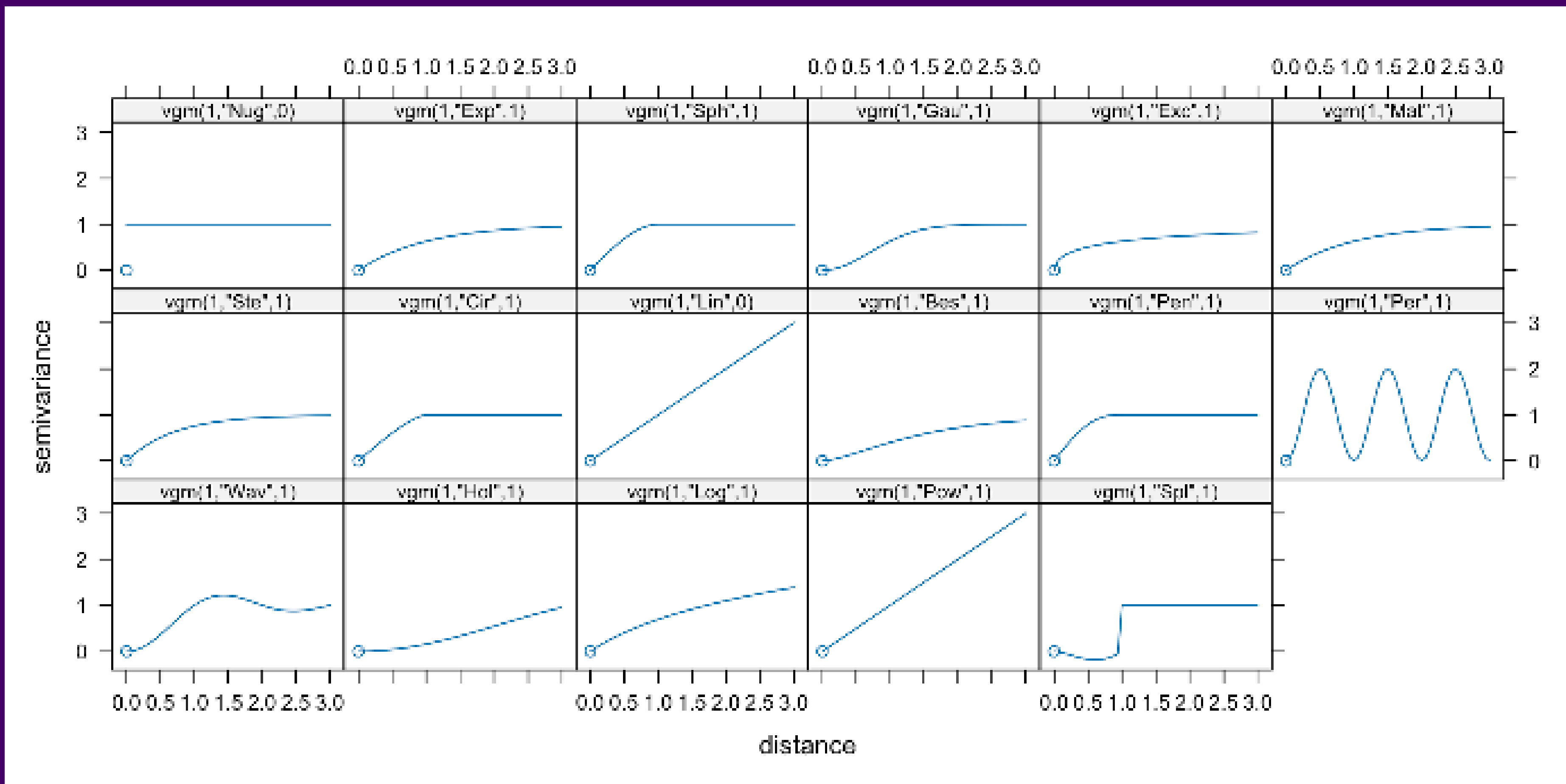
- Removing any **spatial trend** in the data (if present). ** Computing the **experimental variogram**, γ, which is a measure of spatial autocorrelation.
- Defining an **experimental variogram model** that best characterizes the spatial autocorrelation in the data.
- Interpolating the surface using the experimental variogram.

(semi)Variograms

- Relationship between observation variance and distance
- **nugget** - the proportion of semivariance that occurs at small distances
- **sill** - the maximum semivariance between pairs of observations
- **range** - the distance at which the **sill** occurs

Variograms

- Empirical vs. fitted.



Universal Kriging

- Assumes that the deterministic part of the process ($\mu(\mathbf{x})$) is now a function of the location $\mathbf{x}$
- Could be the location or some other attribute
- Now y is a function of some aspect of x

Co-Kriging

- relies on autocorrelation in $\epsilon_1(\mathbf{x})$ for z_1 AND cross correlation with other variables ($z_{2\ldots j}$)
- Extending the ordinary kriging model gives:

$$z_1(\mathbf{x}) = \mu_1 + \epsilon_1(\mathbf{x})$$
$$z_2(\mathbf{x}) = \mu_2 + \epsilon_2(\mathbf{x})$$

- Note that there is autocorrelation within both z_1 and z_2 (because of the ϵ) and cross-correlation (because of the location, $\mathbf{x}$)
- Not required that all variables are measured at exactly the same points

Why Krig?

- Allows empirical estimate of autocorrelation
- Incorporate actual spatial structure
- Increased accuracy
- Uncertainty estimates

Why Not Krig?

- Computationally intensive
- Smoothing not guarenteed
- Variograms (somewhat) subjective